SUDHANSHU SRIVASTAVA

Machine Learning Engineer | AI Researcher | Data Scientist
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PROFESSIONAL SUMMARY

AI/ML researcher with 7+ years of experience in reinforcement learning, multi-agent systems, causal inference, and stochastic optimization. Designed 4 novel deep RL frameworks for manufacturing that match ILP-optimal solutions at 8,000x faster inference, outperform optimization baselines by up to 53 percentage points under disruption, and cut production scheduling costs by 20–30%. 9 publications across IISE Transactions, IJPR, Expert Systems with Applications, and Nature Scientific Reports. Validated all systems on real-world data from a Fortune 500 computer server manufacturer (NSF + IBM GOALI project).

TECHNICAL SKILLS

Machine Learning & Deep Learning: Reinforcement Learning (Multi-Agent, Hierarchical, Actor-Critic, PPO, MAPPO), Causal Discovery (DAG Learning, Vec2DAG), GANs, XGBoost, Random Forest, Gradient Boosting, Bayesian Optimization, CNNs, LSTMs, Transformers, Transfer Learning, NLP, Computer Vision

Optimization & Operations Research: Integer Linear Programming (CPLEX, Gurobi), Lexicographic Optimization, Stochastic Programming, Cost-Sensitive Learning, Warm-Start Hybrid RL+ILP, Predictive Analytics, Simulation Modeling

Languages & Frameworks: Python, R, SQL, PyTorch, TensorFlow, Stable-Baselines3, OpenAI Gym, scikit-learn, pandas, NumPy, SciPy, Matplotlib, Seaborn, NLTK, spaCy

Tools & Infrastructure: Git, GitHub, Linux, Docker, Jupyter, NVIDIA A6000 GPU, Weights & Biases, LaTeX, ArcGIS, SNAP, ERDAS Imagine, Tableau, Microsoft Excel

Domains: Semiconductor Manufacturing, Production Scheduling, Defect Root-Cause Analysis, Supply Chain Optimization, Demand Forecasting, Customer Segmentation, Remote Sensing, Geospatial Analytics

PROFESSIONAL EXPERIENCE

ML Research Scientist — Graduate Researcher (NSF + IBM GOALI) Aug 2024 – Present
University of Louisville *Louisville, KY*

- Built **CF-MARL-SMV2D**, a multi-agent causal RL framework for semiconductor defect root-cause analysis on **5,173 real-world records (29 features)**; reduced structural Hamming distance by **62.5%** over the best baseline, achieving **0.93 TPR**, **0.04 FDR**, and **96.1% downstream defect prediction accuracy**, outperforming 8 SOTA methods (NOTEARS, DAGMA, ALIAS, etc.)
- Designed **LGRGANN + WBO-CIMP**, a GAN-augmented label reconstruction + Bayesian optimization pipeline for defect disposition under **20%+ missing labels** and **5–10x asymmetric misclassification costs**; improved Scrap-class F1 from 0.76 to **0.92 (+21%)** and reduced expected cost by **25.3%** (validated across n=50 runs: 10 seeds × 5 temporal folds)
- Engineered **LHAC**, a lexicographic dual-agent deep RL scheduler for server-to-cell assignment across **4 test banks × 14 cells**; achieved **98.5% completion** / **1.4% tardiness** deterministically, and **outperformed GA+CPLEX by up to 53 percentage points** under stochastic arrivals, processing noise, and 20% cell disruptions
- Developed **HMARL**, a hierarchical multi-agent RL system for distributed production coordination; achieved **8.08% cost reduction** (within 0.44% of ILP) with **<0.01s inference vs. 83.4s (8,340x speedup)**; scaled to **500 orders at 87.5% on-time delivery** in <8s where ILP becomes computationally intractable
- Built a **Hybrid RL+CPLEX warm-start pipeline** feeding trained LHAC policies as feasible initial solutions to a CPLEX MIP solver, recovering additional server placements on capacity-stressed instances where RL-only completion dropped to 93%

Senior Research Associate Aug 2023 – Dec 2023
Indian Institute of Management (IIM) Ahmedabad *Gujarat, India*

- Developed ML/DL pricing optimization and demand forecasting models on live transactional data for **Nykaa ($7B+ market cap)**, India's largest beauty e-commerce platform, under Prof. Indranil Bose
- Engineered customer segmentation pipelines using clustering and classification on **500K+ transaction records**, informing targeted marketing strategy that improved campaign targeting precision

AI Consultant Feb 2024 – Jun 2024
Hope and Dreams Trust (Education NGO) *Uttar Pradesh, India*

- Built NLP-driven predictive analytics pipelines for scholarship targeting and student outreach across **1,000+ beneficiaries**; developed automated reporting tools that reduced manual data processing time by **40%**

Assistant Professor, School of Computer Science Aug 2019 – Jul 2023
UPES Dehradun *Uttarakhand, India*

- Designed and delivered ML, Deep Learning, NLP, and Compiler Design curricula across 8 semesters to **400+ students**; supervised 8 capstone projects, **2 resulting in peer-reviewed conference publications**
- Built a deep-learning Vocal Psychiatric Simulator for public speaking anxiety treatment, **adopted by the university counseling center** (published at ICIVC 2021); filed **3 Government of India software copyrights** for original ML tools

Project Scientist Aug 2018 – May 2019
Indian Institute of Remote Sensing (IIRS), ISRO *Uttarakhand, India*

- Developed a SAR-optical data fusion pipeline (Sentinel-1 + Sentinel-2) for human settlement detection; improved class separability from TD=1.025 to **TD=1.93 (+88%)**, achieving **92% classification accuracy** (κ=0.90) on government-funded geospatial project

System Analyst Jun 2016 – Jul 2017
Maven General Contracting L.L.C. *Abu Dhabi, UAE*

- Built data-driven procurement and cost forecasting models for construction operations across **10+ active sites**; reduced cost variance by **18%** and developed automated analytics dashboards improving project tracking efficiency

EDUCATION

PhD, Industrial & Systems Engineering Aug 2024 – Dec 2026 (Expected)
University of Louisville | Thesis: Reinforcement Learning for Distributed Manufacturing | NSF + IBM GOALI Funded

MBA, Operations Research Jan 2023 – Jan 2025
UPES Dehradun

MTech, Computer Science & Engineering Jul 2017 – May 2019
Indian Institute of Technology (IIT) Dhanbad | GATE 96th Percentile | Government Fellowship Recipient

BE, Computer Science Jul 2012 – Aug 2016
Birla Institute of Technology (BIT) Mesra, Ranchi

SELECTED PUBLICATIONS (9 TOTAL)

- S. Srivastava et al. “Counterfactual Multi-Agent RL for Causal Defect Analysis in Manufacturing.” Intl. Journal of Production Research (under review)
- S. Srivastava et al. “Defect Disposition Prediction: Cost-Sensitive Bayesian Optimization with Label Reconstruction.” Expert Systems with Applications (under review)
- S. Srivastava. “Human Settlement Detection Using Image Fusion.” Nature Scientific Reports (under review)
- S. Srivastava et al. “Hierarchical MARL for Distributed Manufacturing.” IISE Annual Conference 2026
- S. Srivastava et al. “Integrated ML for Semiconductor Defect Analysis.” IISE Annual Conference 2025

  + 4 additional publications (IISE 2026, ICAIA 2023, ICDSA 2022, ICIVC 2021)

AWARDS & LEADERSHIP

- Outstanding Service Award, ISE Dept., University of Louisville (2026) — top 2 of all graduate students
- ASQ Louisville Section 912 Scholarship Award (2025) — for research potential in Quality Engineering
- NSF Student Travel Award, IISE Annual Conference (2025) — NSF Award CMMI-2511912
- Runner-up, UofL Graduate School Three Minute Thesis Competition (2025)
- President, Indian Graduate Student Association (IGSA), UofL (2025–2026) — led 9-member board, organized 7+ events, won Outstanding New Student Organization Award at 27th Annual UofL Student Awards
- GATE Fellowship, Government of India (2017–2019) — 96th percentile; fully-funded MTech at IIT Dhanbad